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Electronics JE 2019

Important Formulas of Basic Electrical Engineering



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IMPORTANT FORMULA

BASIC CONCEPTS OF DC CIRCUIT

1. RESISTANCE

- $R = \rho \frac{l}{A}$

Where

ρ = specific resistance or resistivity of the material

l = length of the conductors

A = cross-sectional area of conductor

- **Conductance (G)**

$G = \frac{1}{R}$.

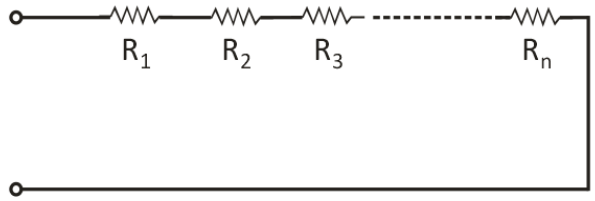
- **Temperature coefficient of resistance**

$\therefore \qquad \alpha = \frac{R_t - R_0}{R_0 . t}$

where R_0 is resistance at 0° C, R_T is resistance at t^o C, and t is temperature rise in ^{o} C.

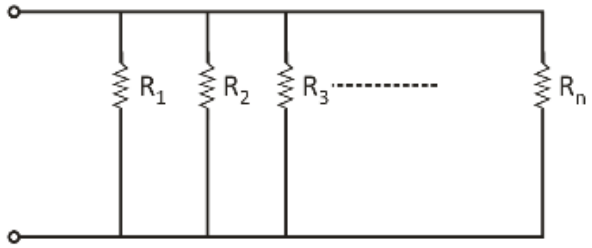
- **Series & Parallel Combination of Resistors:**

(i) Series Combination:



$R_{eq} = R_1 + R_2 + R_3 \ldots . R_n$

(ii) Parallel Combination:



$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \ldots \frac{1}{R_n}$

2. INDUCTANCE

- $L = -\frac{v}{\frac{di}{dt}}$

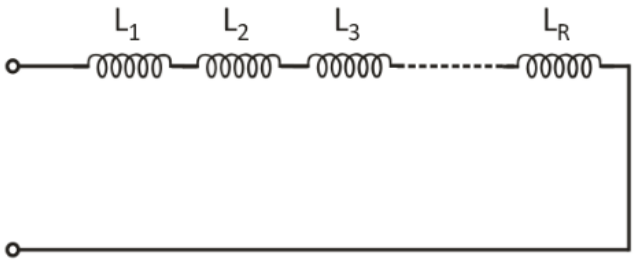
Or we can say

$v = -L \frac{di}{dt}$

Where minus sign indicates the opposing nature of the emf induced in the circuit due to Lenz law.

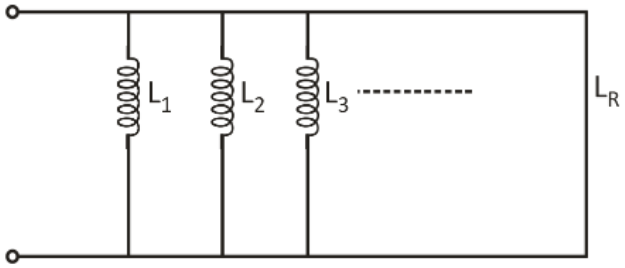
- **Series & Parallel Combination of Inductors:**

(i) Series Combination:



$$L_{eq} = L_1 + L_2 + ... + L_R$$

(ii) Parallel Combination:



$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} .... \frac{1}{L_R}$$

3. CAPACITANCE

- the formula for the capacitance is given as:

$$C = \frac{Q}{V}$$

For Parallel plate capacitor,

$$C = \frac{\in A}{d}$$

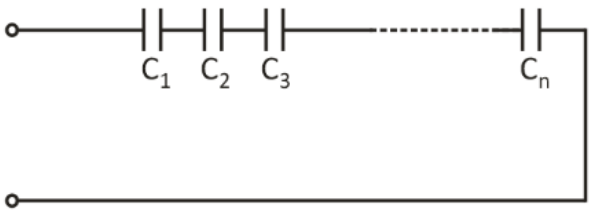
Where $\in$ is the permittivity of free space

A is the cross-sectional area of the plates

d is the distance between the plates

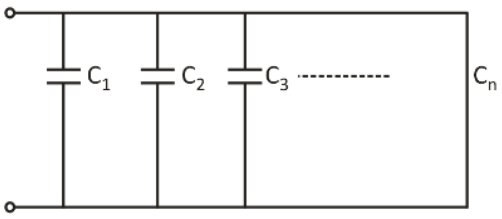
- **Series & Parallel Combination of Capacitors:**

(i) Capacitance in series



$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} ..... \frac{1}{C_n}$$

(ii) Capacitance in parallel



$$C_{eq} = C_1 + C_2 + ...C_n$$

4. ELECTRIC CHARGE

Q=ne
Where, Q= charge
e = 1.6 * 10^{-9} C
n= 1,2,3……..

5. ELECTRIC CURRENT

$$I(\text{Ampere}) = \frac{\text{coulomb}}{\text{second}}$$
$$= \frac{\Delta q}{\Delta t}.$$

One coulomb is equivalent to the charge of 6.28 × 10^{18} electrons.

6. ELECTRICAL POWER

Electrical power is expressed in terms of watts (W) and is given by

$$W = E \times I$$
$$= I^2 R$$
$$= \frac{E^2}{R}$$

Power is also expressed in terms of kW (kilowatt) (= 1000 W) or MW (megawatt) which is 1000 kW or 1000,000 W.

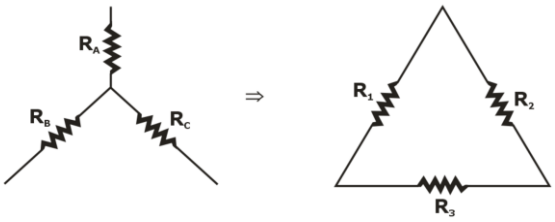
7. ELECTRICAL ENERGY

E=P*T

Electrical Energy is expressed in terms of kilowatt hours(kWh). Thus
1 kWh = 1 kW × 1 hour
= 1000 watt – hours
= 1000 × 60 × 60 watt- Sec

8. STAR – DELTA/ DELTA – STAR TRANSFORMATION

- **Star to Delta Conversion:**

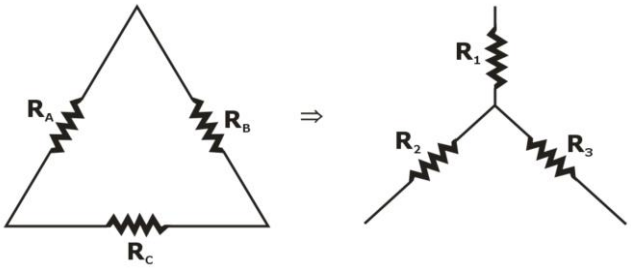


$$R_1 = R_A + R_B + \frac{R_A R_B}{R_C}$$

$$R_2 = R_A + R_C + \frac{R_A R_C}{R_B}$$

$$R_3 = R_B + R_C + \frac{R_B R_C}{R_A}$$

- **Delta to Star Conversion:**



$$R_1 = \frac{R_A R_B}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_B R_C}{R_A + R_B + R_C}$$

NETWORK LAWS & THEOREMS

9. OHM'S LAW

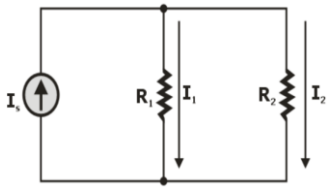
$$\text{Current} = \frac{\text{Applied voltage}}{\text{Resistance of the circuit}} = \frac{V}{R}$$

$$V = IR = \text{Current} \times \text{Resistance}$$

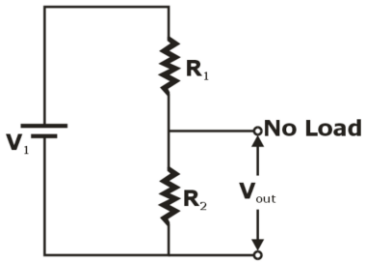
10. CURRENT DIVISION RULE

$$I_1 = \frac{R_2}{R_1 + R_2} I_S$$

$$I_2 = \frac{R_1}{R_1 + R_2} I_S$$



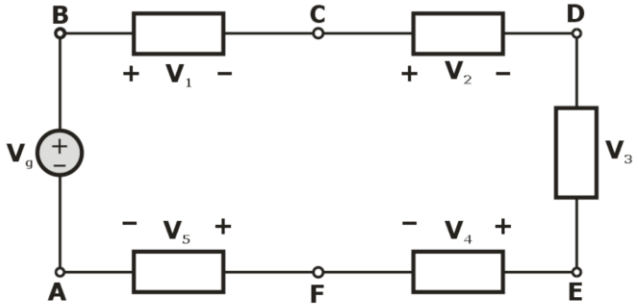
11. V0LTAGE DIVISION RULE



$$V_{out} = V_1 \frac{IR_2}{I(R_1 + R_2)} = \frac{V_1 R_2}{(R_1 + R_2)}$$

12. KIRCHHOFF'S LAWS

- **Kirchhoff's Voltage Law (KVL):**



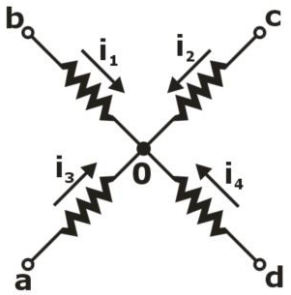
According to the Kirchhoff's voltage law:

$V_g - V_1 - V_2 - V_3 - V_4 - V_5 = 0$

Mathematically KVL can be written as

$$\Sigma V = 0$$

- **Kirchhoff's Current Law (KCL):**



According to KCL at node O, we have

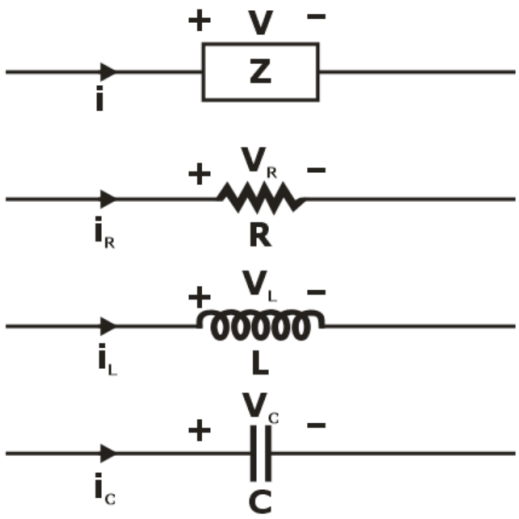
$i_1 + i_2 - (i_3 + i_4) = 0$

Or $i_1 + i_2 = i_3 + i_4$

Mathematically KCL can be written as

$\Sigma i = 0$

13. RELATION OF CURRENT AND VOLTAGE THROUGH DIFFERENT PASSIVE ELEMENT



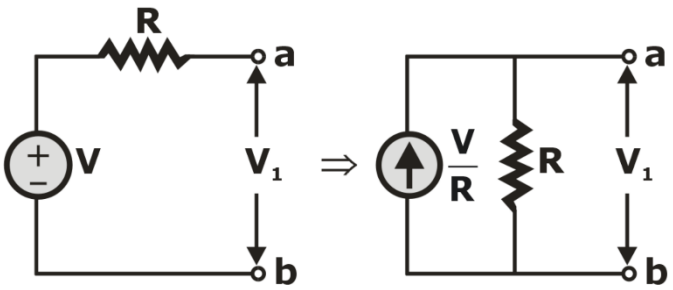
$$V_R(t) = R\, i_R(t)$$

$$V_L(t) = L\frac{di_L(t)}{dt}$$

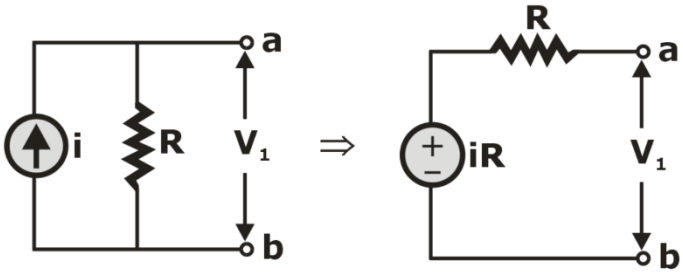
$$V_C(t) = \frac{1}{C}\int_{-\infty}^{t} i_c(t)\,dt$$

14. SOURCE TRANSFORMATION

(i) Voltage Source transformed into Current Source:

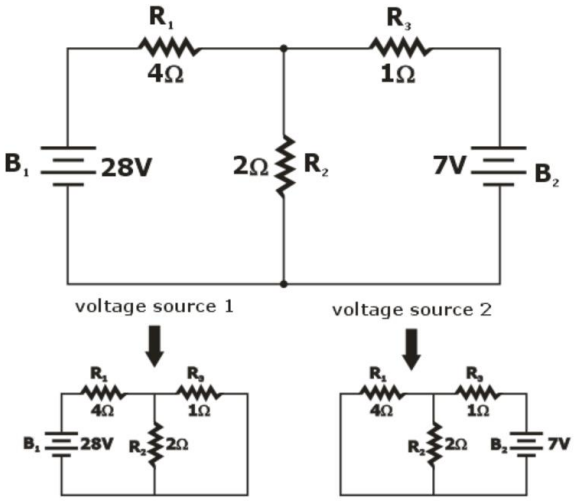


(ii) Current Source transformed into Voltage source:

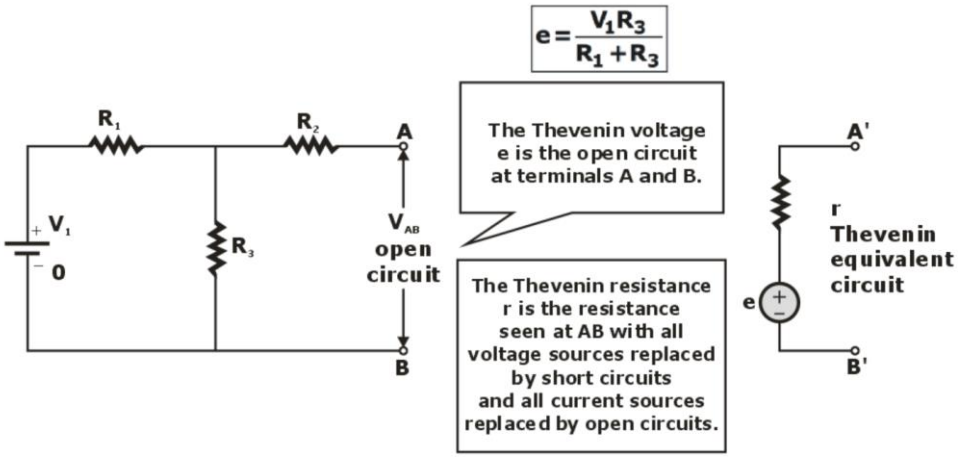


15. SUPERPOSITION THEOREM

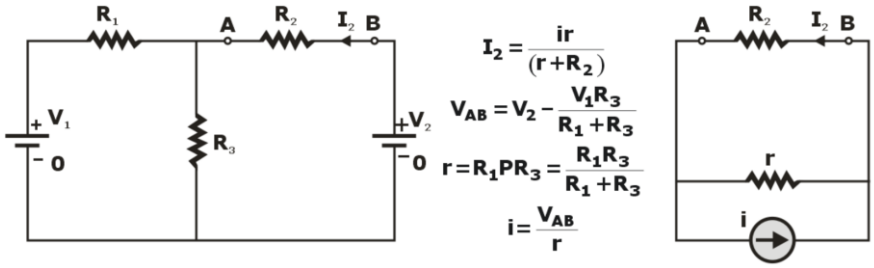
The Super position theorem is a way to determine the currents and voltages present in a circuit that has multiple sources (considering one source at a time)



16. THEVENIN'S THEOREM



17. NORTON'S THEOREM



18. MAXIMUM POWER TRANSFER THEOREM

- As per the maximum power transfer theorem, R_{TH} is the load resistance of the network, i.e., $R_L = R_{TH}$ that allows maximum power transfer.

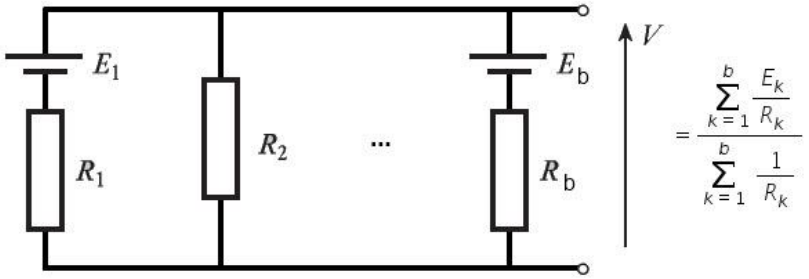
- Maximum Power Transfer is calculated by the equation shown below

$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$

19. TELLEGAN'S THEOREM

$$\sum_{k=1}^{n} P_k = V_k \times I_k = 0$$

20. MILLMAN'S THEOREM



21. RECIPROCITY THEOREM

The theorem states that in a linear passive bilateral network, the excitation source and its corresponding response can be interchanged

AC CIRCUIT

22. REACTANCE

Component	Reactance
Resistor	$X_R = R$
Capacitor	$X_C = 1/\omega C$
Inductor	$X_L = \omega L$

Component	Reactance
resistor	$X_R = R$
capacitor	$X_C = 1/\omega C$
inductor	$X_L = \omega L$

23. IMPEDENCE FOR RLC SERIES CIRCUIT

$$Z_{RL-series} = \sqrt{R^2 + X_L^2}$$

$$Z_{RC-series} = \sqrt{R^2 + X_C^2}$$

$$Z_{LC-series} = \sqrt{(X_L - X_C)^2}$$

$$Z_{RLC-series} = \sqrt{R^2 + (X_L - X_C)^2}$$

$$X_L = 2\pi fL$$

$$X_C = \frac{1}{2\pi fC}$$

24. IMPEDENCE FOR PARALLEL RLC CIRCUIT

$$\frac{1}{Z_{RL-parallel}} = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_L}\right)^2}$$

$$\frac{1}{Z_{RC-parallel}} = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_C}\right)^2}$$

$$\frac{1}{Z_{LC-parallel}} = \sqrt{\left(\frac{1}{X_L} - \frac{1}{X_C}\right)^2}$$

$$\frac{1}{Z_{RLC-parallel}} = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_L} - \frac{1}{X_C}\right)^2}$$

25. OHM'S LAW FOR AC

$$E = IZ \qquad I = \frac{E}{Z} \qquad Z = \frac{E}{I}$$

Where,

E = Voltage in volt

I = Current in amperes (amps)

Z = Impedance in ohms

26. SERIES AND PARALLEL IMPEDANCES

$$Z_{series} = Z_1 + Z_2 + \ldots Z_n$$

$$Z_{parallel} = \frac{1}{\frac{1}{Z_1} + \frac{1}{Z_2} + \ldots \frac{1}{Z_n}}$$

27. RESONANCE

$$f_{resonant} = \frac{1}{2\pi\sqrt{LC}}$$

NOTE: This equation applies to a non-resistive LC circuit. In circuits containing resistance as well as inductance and capacitance, this equation applies only to series configurations and to parallel configurations where R is very small.

28. AC POWER

P = true power $\quad P = I^2R \quad P = \frac{E^2}{R}$

*Measured in units of **Watts***

Q = reactive power $\quad Q = I^2X \quad Q = \frac{E^2}{X}$

*Measured in units of **Volt-Amps-Reactive (VAR)***

S = apparent power $\quad S = I^2Z \quad S = \frac{E^2}{Z} \quad S = IE$

*Measured in units of **Volt-Amps (VA)***

$$P = (1E)(\text{power factor})$$

$$S = \sqrt{P^2 + Q^2}$$

$$\text{Power factor} = \cos(\text{Z phase angle})$$

29. **TIME CONSTANT**

FOR RC CIRCUIT = RC

FOR RL CIRCUIT = L/R

TWO PORT NETWORK

30. Z OR IMPEDENCE MODEL:

$$\begin{aligned} V_1 &= Z_{11}I_1 + Z_{12}I_2 \\ V_2 &= Z_{21}I_1 + Z_{22}I_2 \end{aligned} \qquad \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \mathbf{Z} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}, \quad Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0}, \quad Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}, \quad Z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0}$$

31. Y OR ADMITTANCE MODEL:

$$\begin{aligned} I_1 &= Y_{11}V_1 + Y_{12}V_2 \\ I_2 &= Y_{21}V_1 + Y_{22}V_2 \end{aligned} \qquad \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \mathbf{Y} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$Y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2=0}, \quad Y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1=0}, \quad Y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0}, \quad Y_{22} = \left.\frac{I_2}{V_2}\right|_{Y_1=0},$$

32. A OR TRANSMISSION MODEL:

$$\begin{aligned} V_1 &= A_{11}V_2 + A_{12}(-I_2) \\ I_1 &= A_{21}V_2 + A_{22}(-I_2) \end{aligned} \qquad \begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} = \mathbf{A} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$A_{11} = \left.\frac{V_1}{V_2}\right|_{I_2=0}, \quad A_{12} = \left.\frac{V_1}{I_2}\right|_{V_2=0}, \quad A_{21} = \left.\frac{I_1}{V_2}\right|_{I_2=0}, \quad A_{22} = \left.\frac{I_1}{I_2}\right|_{V_2=0}$$

33. H OR HYBRID MODEL:

$$\begin{aligned} V_1 &= H_{11}I_1 + H_{12}V_2 \\ I_2 &= H_{21}I_1 + H_{22}V_2 \end{aligned} \qquad \begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix} = \mathbf{H} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$H_{11} = \left.\frac{V_1}{I_1}\right|_{V_2=0}, \quad H_{12} = \left.\frac{V_1}{V_2}\right|_{I_1=0}, \quad H_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0}, \quad H_{22} = \left.\frac{I_2}{V_2}\right|_{I_1=0}$$

34. Z-MODEL AND Y-MODEL ABOVE CAN BE WRITTEN IN MATRIX FORM:

$V = ZI$, $I = YV$, $Y = Z^{-1}$

35. COMBINATIONS OF TWO-PORT MODELS

- Series connection of two 2-port networks: $Z = Z_1 + Z_2$
- Parallel connection of two 2-port networks: $Y = Y_1 + Y_2$
- Cascade connection of two 2-port networks: $A = A_1.A_2$